



Mind2Motion - EEG Motor Movement Classification

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Introduction & Motivation

Our goal is to build a model using python that can classify different motor movements from EEG signals. Specifically, left vs. right hand movements and both fists vs. both feet movements. Brain-computer interfaces rely on detecting these neural patterns to enable hands-free control of devices. Reliable classification of motor-related EEG signals could support applications such as assistive technologies or alternative control systems.

Dataset Description

Source	PhysioNet EEG Motor Movement/Imagery Dataset
Subjects	109 Volunteers
Recordings and Channels	64 EEG electrodes, 20000 samples / 125 seconds $\approx$ 160 Hz sampling rate
Experimental Runs (14) We are only focusing on the real motor movement (Task 1 & Task 3)	- Baseline, eyes open - Baseline, eyes closed - Task 1 (open and close left or right fist) - Task 2 (imagine opening and closing left or right fist) - Task 3 (open and close both fists or both feet) - Task 4 (imagine opening and closing both fists or both feet) - Repeat Tasks 1-4 two more times
Annotations	- T0 corresponds to rest - T1 corresponds to onset of motion of the left fist and both fists - T2 corresponds to onset of motion of the right fist and both feet

Related Work



Class Readings



BCI Tutorial, De Sa Lab (BCI_TUTORIAL.pdf)

- Explains fundamental challenges of EEG-based BCI
- EEG signals vary significantly across individuals due to differences in brain anatomy and neural activity

Brain-Computer Interfaces: A Gentle Introduction (gentleintro.pdf)

- EEG oscillations are divided into frequency bands (delta, theta, alpha, beta, gamma)
- Motor activity primarily affects the mu and beta frequency bands

Common Spatial Patterns (CSP.pdf)

- A widely used feature extraction method for motor imagery EEG signals
- Finds spatial filters that maximize and minimize variance for classes

Recent Paper in Literature



We explored a scientific journal from MDPI focusing on Motor Imagery Analysis from Extensive EEG Data Representations. We learned that:

- Recent motor imagery BCI research commonly focuses on 3 central EEG electrodes located over the sensorimotor cortex: C3, Cz, and C4.
- These electrodes correspond to brain regions for voluntary motor control
- Many motor imagery studies extract features from C3, Cz, and C4 to classify movements.
- When building our model we should use C3 and C4 to capture motor cortex activity

Source: <https://www.mdpi.com/1424-8220/22/16/6093>

Methods



Baseline Model: Hand Movements



Approach:

- Used data from Subject 1 (S001) only
- Focused only on hand movement runs (R03, R04, R07, R08, R11, R12)
- Extracted features from EEG epochs (cut continuous EEG into small windows around each movement trial)
- Split data into 70% training 30% testing split

Model:

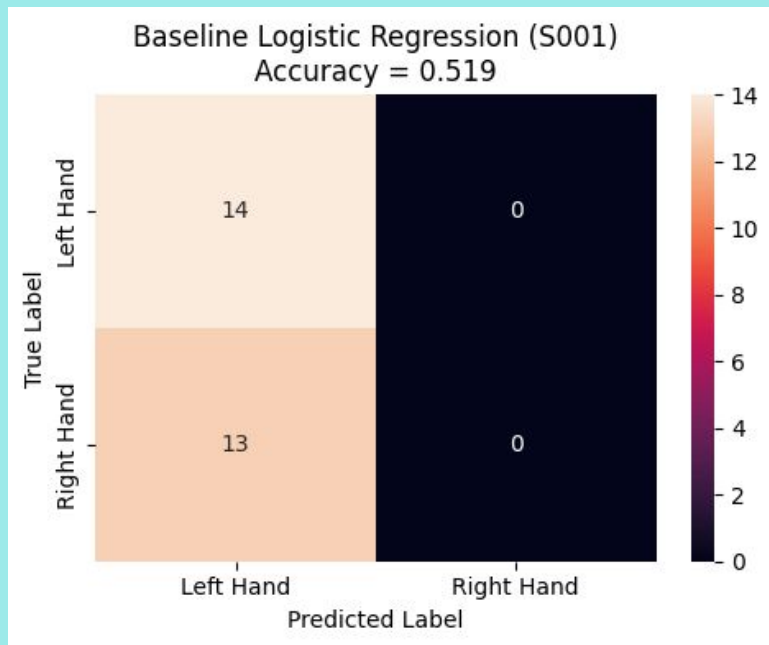
- Logistic Regression classifier, implemented using scikit-learn

Evaluation:

- Accuracy score
- Confusion matrix



Why the Baseline Model Failed



Issue 1: Very Small Dataset

- Initial experiment used runs from just Subject 1
→ 90 movement trials → limited training data

Issue 2: EEG Noise

- EEG signals are highly noisy
- Simple features struggle to capture motor patterns

Issue 3: Model Bias

- Classifier predicted mostly left-hand movements

Ideas for Improvement



More Data:

- Include multiple subjects to expand dataset
- 6 runs per subject → 15 movement trials per run → more training data

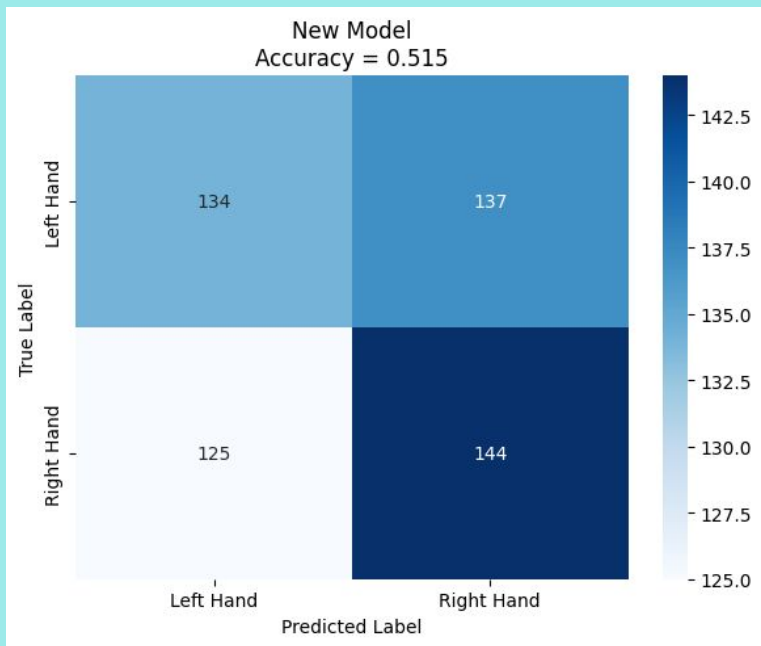
Extract Meaningful EEG Features:

- Focus on electrodes we learned relate to motor movement: C3 and C4
- Analyze EEG frequency bands primarily affected by movement
 - Mu band (8-12 Hz) → decreases during movement
 - Beta band (13-30 Hz) → modulates during movement

Improve Feature Scaling:

- Apply StandardScaler so logistic regression treats all features equally

After Applying Improvements



Model Improvements:

- Expanded dataset to 10 subjects (S001-S010)
- Extracted C3-C4 motor cortex features
- Extracted movement-related EEG features from Mu (8-13 Hz) and Beta (13-30 Hz) bands
- Applied StandardScaler
- Trained Logistic Regression

Problem Observed:

- After increasing dataset and using motor cortex features → Model accuracy still $\approx 52\%$ (essentially random guessing)

Next Steps



Seeking Feedback:

- To improve our model, we sought advice on Piazza from classmates and an instructor. These discussions brought to our attention a key challenge in EEG classification as well as how we can address the issue

EEG Classification Challenge: Inter-Subject Variability

- As we learned from the BCI Tutorial paper, EEG signals vary significantly between individuals due to:
 - Differences in skull thickness
 - Differences in cortical folding
 - Differences in neural activity
 - Small variations in electrode placement
- How to address? → Use subject-dependent models

Final Model: Subject-Dependent



Approach:

- Trained a separate model for each subject
- Selected C3 and C4 electrodes from EEG epochs
- Applied Common Spatial Patterns with $n_components=2$ after selecting C3 and C4 to enhance the differences between classes
- Scaled features using StandardScaler
- Split data into 70% training 30% testing split for each subject

Model:

- Logistic Regression classifier

Evaluation:

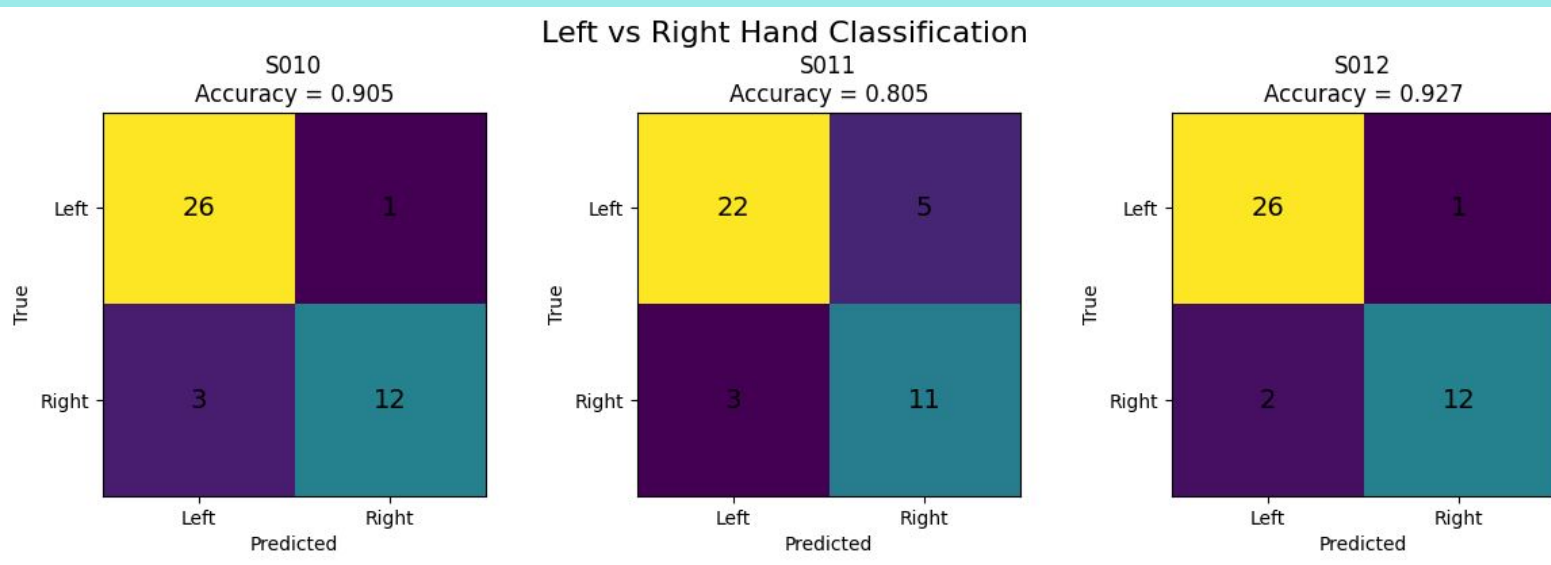
- Accuracy score per subject
- Confusion matrix per subject



Final Model Results



Left vs Right Confusion Matrices



Fists Vs Feet Final Model



Approach:

- Still subject-dependent
- Now focused on both hands vs. both feet runs (R05, R06, R09, R10, R13, R14)
- CSP is applied to all relevant electrodes and it automatically finds the spatial patterns that maximize class differences (no longer pre-selecting only C3 and C4)

Model:

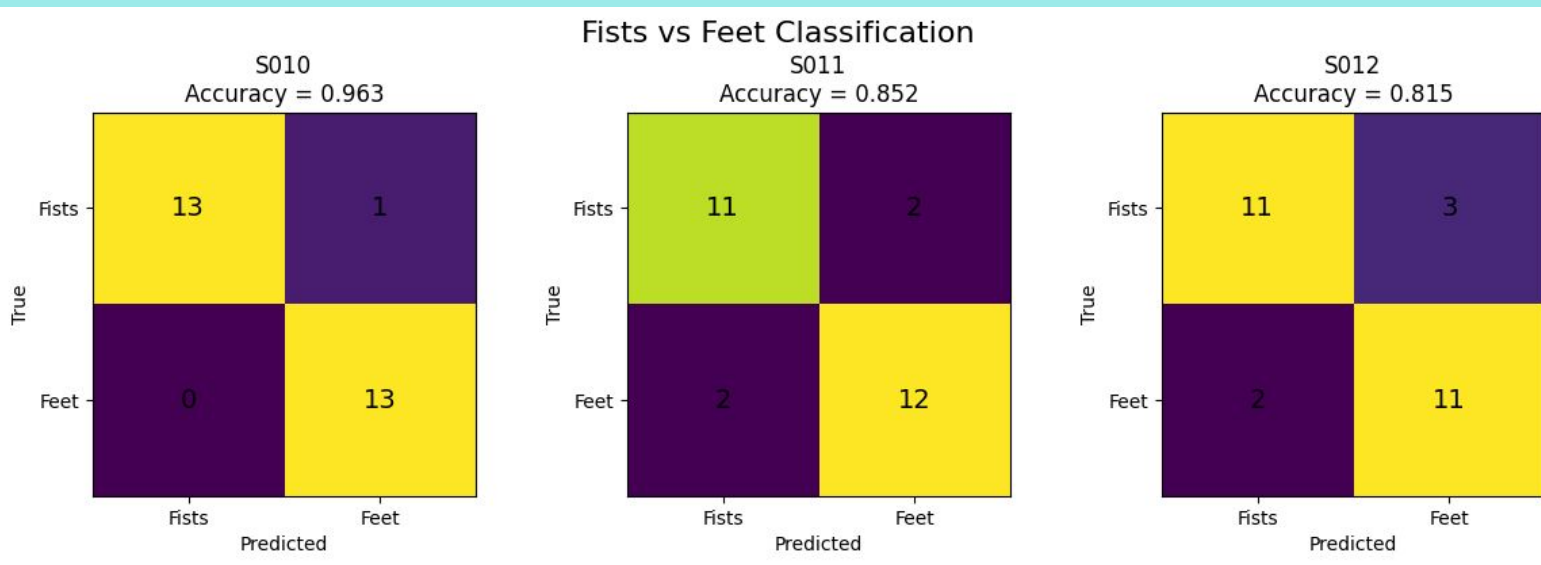
- Logistic Regression classifier

Evaluation:

- Accuracy score per subject
- Confusion matrix per subject



Fists vs Feet Confusion Matrices



Discussion

The background is a vibrant teal color with large, soft, light blue wavy shapes that create a sense of depth and movement. Scattered across this background are several stylized, cartoonish icons. There are four yellow four-pointed stars with purple outlines, two yellow lightning bolts with purple outlines, and three small purple circles with white outlines. Additionally, there are three small yellow circles with purple outlines, each containing a small white dot. The overall aesthetic is playful and modern.

What We Learned



Inter-Subject Variability is a Major Challenge:

- EEG signals vary significantly across individuals
- It is difficult to build an effective classification model that generalizes across many subjects

Subject-Dependent Models Perform Better:

- Training a separate model per subject significantly improved performance
- This suggests that BCI often require personalized calibration

EEG Data is Extremely Noisy:

- Raw EEG signals contain artifacts and noise
- Effective feature extraction such as CSP is critical for capturing motor imagery patterns

Next Time & Improvements



Extension 1: Try Different, More Advanced Models

- Explore models beyond just a logistic regression classifier such as SVM, Random Forest, or neural networks
- More complex models may be better at capturing the non-linear patterns in EEG signals that our model was unable to detect

Extension 2: Reduce Need for Subject Calibration

- Attempt to build model that does not require initial EEG recordings from a user to reliably classify their brain signals
- Our model depends on subject-specific training data, limiting real-world usability
- Future work could explore domain adaptation to generalize across subjects



Thank You!

